Quintic points on thirty remaining modular curves

A consolidated proof and reproducible certificates

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Abstract

We prove finiteness of degree-five closed points over $\mathbb{Q}$ for the thirty levels left open in Section 7 of Derickx–Hwang–Jeon–Orlić [1]. None admits a degree-five function over $\mathbb{Q}$. The proof uses exhaustive finite-field divisor calculations and one local lifting lemma. This note separates the mathematical deductions from the computations. All necessary Magma code and verification output accompany it.

1 Result and reading guide

Theorem 1. *For every level in Table 1, $X_0(N)$ has only finitely many closed points of degree exactly five over $\mathbb{Q}$, and has no function of degree exactly five over $\mathbb{Q}$.*

Finiteness does not mean emptiness: we do not enumerate the quintic points. A degree-five point means one closed point with residue field of degree five, not an arbitrary effective divisor of degree five. No step assumes BSD or equality between analytic and algebraic rank.

There are two computational ideas: *enumerate all degree-five divisor classes over a small finite field*, and *lift a surviving family when its tangent space has the expected dimension.* Three routes organize their application:

- R: direct reduction (rank zero), or no surviving subgroup coset;
- Q: a rank-zero quotient labels all possible abelian translates;
- T: tangent spaces force surviving translates into a known family.

The function column records the separate obstruction to a degree-five function. The double-cover argument is a one-line genus inequality. Local pencils use the same lifting lemma as route T.

Table 1: All thirty cases and their proofs.

N	g	Finiteness route	Degree-five function obstruction
74	8	T	Local pencil, two-prime sieve
77	7	T	Local pencil
79	6	Q	Quotient sieve
82	9	T	Published $\mathbb{Q}$-gonality 6
83	7	Q	Double cover
85	7	R (empty cosets)	Local pencil
86	10	T	$W_5^1(\mathbb{F}_7) = \varnothing$
88	9	R (empty cosets)	Local pencil
89	7	Q	Double cover

N	g	Finiteness route	Degree-five function obstruction
92	10	R (empty cosets)	Double cover
96	9	R (rank zero)	Rank-zero lifting
98	7	R (rank zero)	Rank-zero lifting, with $X_0(49)$
99	9	T	$W_5^1(\mathbb{F}_5) = \varnothing$
100	7	R (rank zero)	Rank-zero lifting
101	8	Q	Double cover
103	8	Q	Quotient sieve
107	9	Q	Double cover
111	11	T	Double cover
118	14	T	Double cover
123	13	T	Double cover
128	9	T	$W_5^1(\mathbb{F}_3) = \varnothing$
131	11	Q	Double cover
141	15	T	Double cover
142	17	T	Double cover
143	13	T	Double cover
145	13	T	Double cover
155	15	T	Double cover
159	17	T	Double cover
167	14	Q	Double cover
191	16	Q	Double cover

2 The common framework

Put $C = X_0(N)$ and $J = \mathrm{Jac}(C)$. A rational cusp identifies $\mathrm{Pic}^5(C)$ with J. Write W_d^r for the locus of degree-d line bundles L with $h^0(L) \geq r + 1$. Since C has a rational point, rational Picard classes are represented by line bundles over $\mathbb{Q}$; there is no Brauer obstruction.

Suppose first that C has no degree-five function over $\mathbb{Q}$. An irreducible rational effective divisor D of degree five must have $h^0(D) = 1$: a nonconstant rational section would have a nonempty rational pole divisor bounded by D, necessarily equal to D. Thus quintic points inject into $W_5^0(\mathbb{Q})$.

Factor J up to $\mathbb{Q}$-isogeny into modular simple factors. Let $\mathcal{A}$ be the sum of the isotypic parts whose central L-value vanishes. Exact modular-symbol central-value tests identify these parts. The nonvanishing theorem for modular abelian varieties [4] gives $\mathrm{rank}(J/\mathcal{A})(\mathbb{Q}) = 0$. We use only this implication: vanishing factors are *possible* positive-rank factors, without claiming their exact ranks.

Mordell–Lang [5] says that the Zariski closure of $W_5^0(\mathbb{Q})$ is a finite union of points and abelian translates. Every positive-dimensional component with dense rational points is defined over $\mathbb{Q}$ and its direction is contained in $\mathcal{A}$, since its image in the rank-zero quotient is finite. It suffices to show that all such components have fixed proper sub-divisors on their $h^0 = 1$ loci. The remaining finitely many classes then give finitely many quintic points.

2.1 What the finite-field enumeration certifies

At each working prime $p \nmid N$, the scripts check that the reduced canonical model is integral, has a rational place, and that its arithmetic genus equals its function-field genus. It is therefore smooth and geometrically integral and represents the good reduction of C.

Every effective degree-five divisor over $\mathbb{F}_p$ is a sum of closed places of degrees at most five. For each partition of five, we enumerate the required multisets of places. This is exhaustive,

with no height bound. The class group calculation gives all of $W_5^0(\mathbb{F}_p)$: over a finite field every rational Picard class is represented by a line bundle.

If $x + B \subset W_5^0$ over $\mathbb{Q}$, its reduction contains the entire coset $\bar{x} + B(\mathbb{F}_p)$ in $W_5^0(\mathbb{F}_p)$, not just reductions of its rational points. This follows from geometric containment and the smooth abelian models at good reduction. One missing element excludes the translate.

2.2 One local lifting lemma

Lemma 2 (Local rigidity)**.** *Let $R = \mathbb{Z}_p$, let Z be a closed subscheme of a scheme smooth over R, and let $U \to Z$ be a morphism with U smooth over R of relative dimension n. At rational special-fiber points $u \mapsto z$, suppose*

$$\dim_{\mathbb{F}_p} T_z Z_{\mathbb{F}_p} = n, \qquad T_u U_{\mathbb{F}_p} \longrightarrow T_z Z_{\mathbb{F}_p} \text{ is an isomorphism.}$$

Then the morphism identifies the completed local rings.

Proof. Lifts of relative cotangent generators give a surjection $R[[x_1, \dots, x_n]] \twoheadrightarrow \widehat{\mathcal{O}}_{Z,z}$. Its composite to $\widehat{\mathcal{O}}_{U,u} \simeq R[[t_1, \dots, t_n]]$ has invertible linear part modulo p, so is an isomorphism by formal inversion. The surjection has zero kernel. Flatness of Z is not assumed. □

We apply this to $Z_S = \bigcap_{s \in S}(W_5^0 - s)$. Prime-to-$p$ rational torsion points on the special fiber lift uniquely to torsion over R. If S lies in both a candidate direction and a known B, the same lifts belong to both. When the direction is already known to equal B, points of $B(\mathbb{F}_p)$ can instead be lifted arbitrarily inside $B(R)$, including p-primary points: both families are stabilized by all of B.

At a class with $h^0(D) = 1$, the annihilator of $T_{[D]}W_5^0$ in $H^0(K_C)$ is $H^0(K_C - D)$. For $h^0(D) \geq 2$, the tangent space is the full ambient space. These are the first-order determinantal equations of W_5^0. The tangent space of Z_S is computed by spanning these annihilators over the coset, in one fixed basis of $H^0(K_C)$.

If the lemma identifies the local branch with a known proper image V, a candidate translate through that residue class is locally contained in V. Properness makes V closed. A nonempty p-adic open subset of the smooth translate is Zariski dense, so the entire translate lies in V.

3 Excluding degree-five functions

3.1 The double-cover inequality: seventeen cases

A degree-two map from C to a genus-h curve and a degree-five map to $\mathbb{P}^1$ generate the function field, since their degrees are coprime. Castelnuovo–Severi gives $g(C) \leq 2h + 4$. Every row below violates this inequality, excluding degree-five functions even geometrically. The script `arithmetic.m` checks the canonical involutions and their invariant dimensions.

N	g	involution	h	N	g	involution	h
83	7	w_{83}	1	89	7	w_{89}	1
92	10	w_{23}	1	101	8	w_{101}	1
107	9	w_{107}	2	111	11	w_{111}	2
118	14	w_{59}	3	123	13	w_{41}	3
131	11	w_{131}	1	141	15	w_{47}	3
142	17	w_{71}	2	143	13	w_{143}	2
145	13	w_{29}	4	155	15	w_{155}	4
159	17	w_{159}	4	167	14	w_{167}	2
191	16	w_{191}	2				

3.2 Empty special loci and local pencils

The $\mathbb{Q}$-gonality of $X_0(82)$ is six [3, Theorem 1.4]. At $86, 99, 128$, respectively at $p = 7, 5, 3$, the numbers of effective degree-five divisors and their distinct classes are equal. Every class has $h^0 = 1$, so $W_5^1(\mathbb{F}_p)$ is empty. Upper semicontinuity excludes a rational degree-five line bundle with two sections. These checks are part of `translates.m`.

For $77, 85, 88$, `local_pencils.m` checks that every special degree-five divisor has $h^0 = 2$, moving degree four, and belongs to $A_4 + C$ for one degree-four pencil class A_4.

N	p	effective divisors	special divisors	$\dim TW_5^1$
77	5	12528	48	1
85	7	28676	64	1
88	3	3928	32	1

The tangent dimension follows from the basepoint-free pencil trick:

$$\dim T_{[D]}W_5^1 = 8 - g + h^0(K_C - D - A_4).$$

The Petri map has source dimension $2(g-4)$ and kernel $H^0(K_C - D - A_4)$, which proves the formula.

These curves have $\mathbb{Q}$-gonality four [3, Theorem 1.2]. A rational tetragonal pencil lifts the unique class A_4. Its reduction cannot acquire an extra section, since adding a rational point would contradict the checked degree-five bound $h^0 \le 2$. The Abel map $C \to W_5^1$, $P \mapsto A_4 + P$, is an immersion. Lemma 2 identifies every relevant formal neighborhood with this family. Its two sections come from A_4, so the added point is fixed. A basepoint-free rational degree-five pencil is impossible.

Level 74 uses the same argument plus a two-prime sieve. Its four rational cusps generate a group of order 513. Here cusp differences are torsion by Manin–Drinfeld, and injectivity at an odd good prime identifies their rational group with its reduction. The quotient by the possible positive part E_{37}^2 has rational torsion order dividing 513. At $p = 3, 5$, the cusp reductions still generate a group of order 513, disjoint from the respective primary groups below. The cusp group therefore maps isomorphically to the rational rank-zero quotient and supplies common labels.

p	primary group used	order	exceptional cusp labels
3	$J(\mathbb{F}_3)[7^\infty]$	49	$(-2, 9, 0),\ (0, -2, -2)$
5	$J(\mathbb{F}_5)[2^\infty]$	256	$(0, -2, 1),\ (1, 6, 0)$

At each prime there is one moving tetragonal class. Outside the two exceptional labels, all eligible special classes have moving degree four and tangent dimension one. The exceptional sets are disjoint *in the same cusp group*; the script checks shared integral representatives, not just inequality of triples. Every rational class is thus covered by local rigidity at one prime. The rational tetragonal pencil again comes from [3]. The file `pencils74.m` includes the arithmetic, cusp identification, and both sieves.

The remaining function exclusions appear below: rank-zero lifting for $96, 98, 100$ and the quotient sieve for $79, 103$.

4 Route R: direct reduction

4.1 Rank zero and the argument for $X_1(32)$

This is where Section 5.5, Proposition 5.5 of Najman–Varivoda [2] applies. All three Jacobians at $96, 98, 100$ have nonzero central L-value, hence rank zero. Reduction on $J(\mathbb{Q})$ is injective

at the odd good primes below. The script `rank_zero.m` enumerates every special degree-five divisor, checks $h^0 = 2$ and moving degree four, and explicitly lifts every moving pencil class by a divisor supported on rational points.

N	p	$\#C(\mathbb{F}_p)$	rational lifts	degree-four pencil classes	$\#W_5^1(\mathbb{F}_p)$
96	7	8	8	2	16
98	3	6	4	1	6
100	3	6	6	3	18

For $96, 100$, write the reduction of a rational special class as $\bar{A}_4 + \bar{P}$. Lift both terms to $\mathbb{Q}$. Injectivity of reduction forces the original class to be $A_4 + P$. Its two sections come from A_4, so P is fixed and no degree-five function results. Thus the proposition applies directly. Rank zero gives finitely many degree-five classes; without degree-five functions each has at most one irreducible quintic representative.

The necessary refinement at 98. The point-lifting hypothesis fails: six finite-field points versus four rational lifts. The proposition cannot be applied unchanged. Use the degree-three degeneracy map $\pi : X_0(98) \to X_0(49) = E$. The code verifies $E(\mathbb{Q}) \simeq \mathbb{Z}/2\mathbb{Z}$ and $E(\mathbb{F}_3) \simeq \mathbb{Z}/4\mathbb{Z}$. Both rational points of E are its rational cusps. The other cusps of E form an orbit of degree six; on $X_0(98)$ there are four rational cusps and two degree-six cusp orbits. The cyclotomic cusp-field description and the order six of 3 modulo 7 show that these other cusps are not $\mathbb{F}_3$-rational. The four supplied points on $X_0(98)$ are checked to be its Atkin–Lehner cusp orbit.

The other two points $\bar{P} \in X_0(98)(\mathbb{F}_3)$ are noncuspidal. Degeneracy maps take precisely cusps to cusps, also at this good prime, so $\pi(\bar{P})$ is a noncuspidal point of $E(\mathbb{F}_3)$ of order four. It is not the reduction of a point of $E(\mathbb{Q})$. In particular $\bar{P}$ cannot lift even as a *rational degree-one Picard class*: pushing such a class forward would give a point of $\mathrm{Pic}^1(E)(\mathbb{Q}) = E(\mathbb{Q})$ reducing to $\pi(\bar{P})$.

Lift the unique degree-four pencil A_4. For a rational special degree-five class L, the rational class $L - A_4$ has degree one. Its reduction must therefore be a rational cusp, and injectivity gives $L = A_4 + P$ with a fixed rational cusp. This proves both assertions at 98.

4.2 Empty cosets: 85, 88, 92

Each has a unique possible positive-rank simple factor, of multiplicity one.

N	$\dim \mathcal{A}$	p	$\#\mathcal{A}(\mathbb{F}_p)$	subgroup order
85	2	3	34	17
88	1	3	7	7
92	1	3	7	7

The indicated prime-primary subgroup of $J(\mathbb{F}_3)$ has that prime order and belongs to $\mathcal{A}(\mathbb{F}_3)$; the complementary Frobenius order is coprime to it. The exhaustive test finds no whole coset inside $W_5^0(\mathbb{F}_3)$. Any positive-dimensional component would have direction $\mathcal{A}$ and would contain such a coset on reduction. Hence none exists. The function exclusions already proved complete the finiteness argument. The file is `empty_cosets.m`.

5 Route Q: nine prime levels

Let $Y = C/\langle w_N \rangle$, $\pi : C \to Y$, $A = \pi^* \mathrm{Jac}(Y)$, and $R = J/A$. Let P be the infinity cusp, $Q = w_N(P)$, $t = [Q - P]$, and $F = P + Q$. The cover is ramified, so pullback on Jacobians is injective. In each row below, A is $\mathbb{Q}$-simple of multiplicity one, and all other factors have nonzero central L-value. The script identifies it by dimension and, at 89, by Frobenius order to distinguish another elliptic factor. Thus R has rank zero.

N	g	$g(Y)$	p	m	$\#A(\mathbb{F}_p)$	surviving indices
79	6	1	3	13	5	$1,2,3,4$
83	7	1	7	41	11	$1,2,3,4,23$
89	7	1	3	22	5	$1,2,3,4$
101	8	1	5	25	7	$1,2,3,4$
103	8	2	5	17	55	$2,3$
107	9	2	3	53	29	$2,3$
131	11	1	3	65	5	$1,2,3,4$
167	14	2	3	83	19	$2,3$
191	16	2	2	95	11	$2,3$

The gcd of $\#R(\mathbb{F}_q)$ over odd primes $q = 3, 5, 7, 11, 13, 17, 19$ is m. Rational torsion injects at these primes, so $\#R(\mathbb{Q})$ divides m. At the working prime, the code constructs $H = A(\mathbb{F}_p)$ and checks

$$\operatorname{ord}(\bar{t}) = m, \qquad \#\langle \bar{t}, H\rangle = m\,\#H.$$

Thus the image of $\bar{t}$ in $R(\mathbb{F}_p)$ has order m, proving $R(\mathbb{Q}) = \langle t \bmod A\rangle$, cyclic of order m. This does not require torsion injectivity at the working prime 2 for 191. At 131, the complement also has 5-torsion, so norms of rational and quadratic places construct the actual $A(\mathbb{F}_3)$ rather than using all 5-torsion.

Every rational degree-five class lies on one of $T_i = [5P] + it + A$, $0 \le i < m$. If $T_i \subset W_5^0$, the full coset $i\bar{t} + H$ must occur. The displayed indices are exactly those passing the test. Simplicity of A implies that every positive-dimensional component with dense rational points is one of these translates.

5.1 Fixed divisors in the surviving families

For $g(Y) = 1$, indices $1, 2, 3, 4$ are respectively

$$\pi^* \operatorname{Pic}^1(Y) + 3P, \quad \pi^* \operatorname{Pic}^2(Y) + P, \quad \pi^* \operatorname{Pic}^2(Y) + Q, \quad \pi^* \operatorname{Pic}^1(Y) + 3Q.$$

For $g(Y) = 2$ only the middle two occur. These identities follow using F or $2F$ as reference pullbacks. The line bundles on Y used here are effective.

The displayed divisors are fixed even when $h^0 > 1$. For an effective divisor D containing no full fiber, $\mathcal{O}_Y$ is a saturated subbundle of $\pi_*\mathcal{O}_C(D)$ whose quotient has degree $\deg D - e$, where $e = g(C) - 2g(Y) + 1$. Saturation follows by intersecting functions from Y with $\mathcal{O}_C(D)$ locally; the degree follows by Euler characteristics. Tensoring with a degree-d line bundle M therefore gives

$$H^0(C, \pi^*M(D)) = H^0(Y, M) \quad \text{if } d + \deg D < e.$$

Use $(d, D) = (1, 3P), (2, P), (2, Q), (1, 3Q)$ as appropriate. The inequality holds in every row. Since P, Q are distinct points of one fiber, none of these divisors contains a full fiber. Every effective representative contains a rational cusp.

The additional cubic family at 83. Let $u^3 + 4u^2 + 2u - 5 = 0$. The canonical point

$$\begin{aligned}[\,&2u^2 + 4u - 4,\ 3u^2 + 5u - 7,\ u^2 + 3u - 3,\ 3u^2 + 3u - 5,\\ &2u,\ -2u^2 - 2u + 2,\ 2\,]\end{aligned}$$

is fixed by w_{83}. The polynomial is irreducible and the fifth coordinate divided by the seventh equals u, so the three conjugates form a rational reduced degree-three divisor D. These exact checks are in `cubic83.m`. The points are ramified, so their multiplicity-one divisor contains no full fiber. The section calculation above shows that D is fixed throughout $\pi^* \operatorname{Pic}^1(Y) + D$. On R, w_{83} acts by -1. For $c = [F + D - 5P] \bmod A$, invariance of $F + D$ gives $2c = 5t$. Since $\#R(\mathbb{Q}) = 41$, $c = 23t$. This accounts for the extra index.

Functions at 79 and 103. The enumeration also records all classes of $W_5^1(\mathbb{F}_p)$ above rational quotient labels. Only labels $2, 3$ occur. Any rational special class therefore lies on T_2 or T_3 over $\mathbb{Q}$, since the quotient labels are distinguished on reduction. Both entire translates have fixed cusps. No degree-five function exists. This test concerns individual special classes, not merely whole cosets, and is part of `quotient_sieve.m`.

All nine cases now follow from Section 2: every surviving positive-dimensional translate has a proper rational fixed divisor, and every function exclusion has been proved.

6 Route T: one family calculation for fifteen levels

Here we force every translate into $V = B + [F_4] + C \subset \mathrm{Pic}^5(C)$. Either $\pi : C \to Y$ is a double cover with $g(Y) = 2$, with $B = \pi^* \mathrm{Jac}(Y)$ and F_4 twice a fiber, or it is a degree-four map to a genus-one curve, with F_4 a fiber. Degree-two line bundles on a genus-two curve and degree-one line bundles on a genus-one curve are effective. Thus $V \subset W_5^0$. It is a proper image of $B \times C$, of expected dimension $b + 1$, where $b = \dim B$.

N	p	quotient generators	b	$\#B(\mathbb{F}_p)$	subgroups tested
74	3	w_{74}	2	21	all order 7
77	5	w_{77}	2	35	all order 7
82	3	w_2, w_{41}	1	6	whole $B(\mathbb{F}_p)$
86	7	w_2, w_{43}	1	8	all cyclic order 8
99	5	w_9, w_{11}	1	10	whole $B(\mathbb{F}_p)$
111	2	w_{111}	2	15	all order 5
118	3	w_2, w_{59}	1	5	all order 5
123	2	w_3, w_{41}	1	3	all order 3 or 5
128	3	w_{128}, $sw_{128}s^{-1}$	1	6	whole $B(\mathbb{F}_p)$
141	2	w_3, w_{47}	1	3	all order 3 or 5
142	3	w_{71}	2	15	all order 5 or 7
143	2	w_{143}	2	15	all order 3
145	3	w_5, w_{29}	1	4	all order 2
155	2	w_5, w_{31}	1	3	all order 3 or 5
159	5	w_3, w_{53}	1	6	all order 3

At 128, s is the rational normalizer action $q \mapsto -q$. All rows are verified by the single routine `translates.m`.

6.1 Why every possible direction is tested

The possible positive simple factors have these profiles; 1^2 denotes a repeated elliptic isotypic part, not a simple surface.

Profile	Levels
1^2	$74, 86, 111, 159$
1	$77, 82, 99, 118, 128, 143$
$1 + 1$ (nonisogenous)	$123, 141, 142, 155$
$1 + 2$ (nonisogenous)	145

For prime-order tests, the Frobenius order of every possible simple constituent is divisible by a tested prime $\ell \neq p$. Every nonzero $\mathbb{Q}$-direction contains a $\mathbb{Q}$-simple subvariety isogenous to such a constituent. Isogeny preserves Frobenius polynomials, so its reduction contains a rational order-ℓ point. The script tests *all* order-ℓ subgroups of $J(\mathbb{F}_p)$, including those from rank-zero factors. Repeated isotypic parts therefore leave no unsearched directions.

At $82, 99, 128$, the unique positive elliptic part equals the known B. The identity is checked on canonical differentials using its newform. The whole $B(\mathbb{F}_p)$ may thus be used, including its p-primary part; this is not a prime-to-p torsion lifting argument.

At 86, the repeated elliptic factor is isogenous to $E_{43} : y^2 + y = x^3 + x^2$. Its 2-division polynomial is irreducible over $\mathbb{Q}$, so it has no rational 2-isogeny. Every elliptic curve in its $\mathbb{Q}$-isogeny class is linked by an odd-degree isogeny: remove full multiplication kernels until the kernel is cyclic; an even cyclic kernel would give a rational order-two subgroup. At 7, E_{43} has eight points and a unique nonzero 2-torsion point, so its group is cyclic of order eight. Odd-degree isogenies preserve this 2-primary group. Every nonzero possible direction therefore contains a rational cyclic order-eight subgroup on reduction. All 192 such subgroups of $J(\mathbb{F}_7)$ are tested.

6.2 The certificate and its lift

Canonical matrices are checked to define the quotient automorphisms and, in degree four, to commute and generate a group of order four. Their invariant differential space has dimension b. The order of $B(\mathbb{F}_p)$ is computed from the quotient Jacobian or its invariant Hecke space. Norms of rational and quadratic places generate a subgroup of that order, constructing the actual $B(\mathbb{F}_p)$.

For *every* tested S and every whole coset $x + S \subset W_5^0(\mathbb{F}_p)$, the certificate checks:

1. $S \subset B(\mathbb{F}_p)$;
2. $x + S \subset B(\mathbb{F}_p) + [F_4] + C(\mathbb{F}_p)$;
3. $\dim T_x Z_S = b + 1$;
4. the differential of $B \times C \to \mathrm{Pic}^5(C)$ has rank $b + 1$ at each relevant rational preimage.

For the last check, the group-trace kernel on canonical differentials annihilates TB; it is evaluated at every rational point of C. The rank and point-count checks also verify the expected smooth reduction of this embedded B.

The complete surviving calculation is below. Counts are subgroup–coset pairs, not necessarily disjoint cosets.

N	effective divisors	$\#W_5^0(\mathbb{F}_p)$	subgroups	surviving cosets	$\dim TZ_S$
74	1340	1316	8	18	3
77	12528	12488	1	40	3
82	2574	2568	1	6	2
86	64384	64384	192	12	2
99	8504	8504	1	6	2
111	328	320	6	12	3
118	3104	3104	6	6	2
123	842	842	$1 + 6$	$6 + 0$	2
128	1812	1812	1	4	2
141	566	566	$1 + 1$	$4 + 0$	2
142	1668	1656	$781 + 57$	$12 + 0$	3
143	332	324	4	20	3
145	1552	1552	15	24	2
155	536	536	$4 + 6$	$4 + 0$	2
159	23476	23476	13	16	2

For two prime orders, the two counts follow their order in the preceding table. All surviving cosets have the displayed tangent dimension.

Let $T \subset W_5^0$ be any positive-dimensional component with dense rational points. Choose a tested subgroup in its reduced direction. The checks give a rational point of $B \times C$ above the reduction of a rational point of T. Lift the subgroup as in Section 2.2. Both T and V lie in Z_S.

Lemma 2 gives $T \subset V$. At 145 this treats the elliptic and simple-surface possibilities together, since both have even Frobenius order at 3. No separate surface calculation is needed.

6.3 Why the known family contributes no quintic points

Project to J/B. The image of V is a translate of the image of C, whose differences generate J/B. In every row $\dim(J/B) > 1$. That curve cannot contain a positive-dimensional abelian translate: such a translate would be an elliptic curve equal to the image, whose differences generate only a one-dimensional variety. The image of T is therefore a point, and $T \subset B + [F_4] + P$ for a fixed geometric point $P \in C$. Each class there has an effective representative containing P.

Every function exclusion has already been proved. If $T \subset W_5^1$, it contributes no irreducible rational quintic divisor. Otherwise, on its dense $h^0 = 1$ open subset, the unique rational representative contains P and its entire Galois orbit. This orbit has degree at most four: more than five is impossible, and degree five would make all classes in the dense open subset equal to the orbit divisor, contradicting $\dim T > 0$. It is thus a proper rational sub-divisor. No positive-dimensional component contributes infinitely many quintic points. Section 2 completes the proof.

7 Reproduction and scope of verification

This folder is independent of the working repository. The computations use Magma V2.29-4. From this folder run

```
python3 run.py --jobs 8
```

Up to 32 independent workers can be requested. To inspect saved output without recalculation, run

```
python3 run.py --check-logs
```

The runner checks all 36 certificates for a success marker and error messages: Magma errors need not produce nonzero exit status. It writes `verification.json` with source and log hashes. Individual scripts run from `code/`, for example

```
magma -b N:=145 translates.m
```

File in `code/`	Purpose	Runs
`arithmetic.m`	Double-cover inequalities	1
`rank_zero.m`	Rank zero, pencil lifts, 98 refinement	3
`empty_cosets.m`	No surviving abelian translates	3
`quotient_sieve.m`	Rational labels and complete cosets	9
`cubic83.m`	Exact fixed cubic divisor	1
`local_pencils.m`	Local W_5^1 test	3
`pencils74.m`	Two-prime exceptional-label sieve	1
`translates.m`	Known families and tangent spaces	15

Helpers `common.m`, `coordinates.m`, `place_cache.m`, and `translation.m` supply exact shared operations and coordinate conversions. The cache accelerates norms; it does not replace exhaustive degree-five enumeration.

The code checks finite groups, divisor classes, Riemann–Roch dimensions, automorphisms and tangent ranks. The deductions use the theorems and proved lemma above; this is not a formal-proof-assistant verification. The published gonality input is used only for rational tetragonal pencils at $74, 77, 85, 88$ and gonality six at 82. The cusp-field description and degeneracy map at 98 are standard modular-curve facts, not numerical searches.

References

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